

You need to know the following topics for test 1 of Signals and Systems

- 1- Find the power or the energy of a signal
- 2- Carry out signal operations such as time scaling and shifting
- 3- Expression of unit steps $u(t)$ and unit impulse $\delta(t)$ and their plots
- 4- Determine if a system is linear, causal, and time-invariant
- 5- Derive the differential equation of a mechanical or an electrical system
- 6- Find the zero-input response
- 7- Find the impulse response $h(t)$
- 8- Convolution
- 9- Find the zero-state response
- 10- Determine if the system is internally stable or not